

Toolchain → gcc GNU compiler collection

IDE ... Eclipse mit ESP-Plugins

[extern int x_add1(int v);] x.h

prototyp angeben

↓
#include "x.h"
int d = x_add1(4);

#include "x.h"

int x_add1(int v) { return v+1; }

RISC-V
Compiler

main.c

x.c

y.c

↓
main.obj

↓
x.obj

↓
y.obj

compile

↳ ASM → elf/obj
assembly

(RISC-V) [Linker

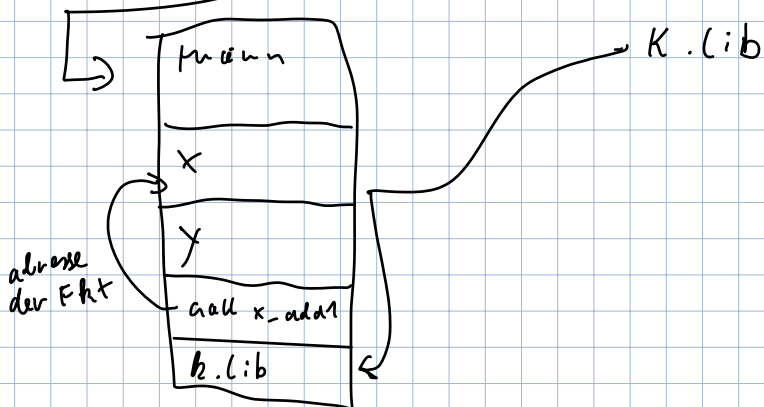
win
*.exe / *.elf

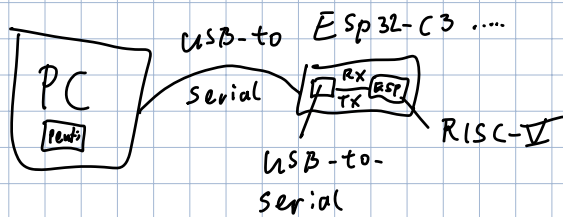
↓
ausführbare Datei

≠
RISC-V

ASM [RISC-V]

call x_add1 → Linker



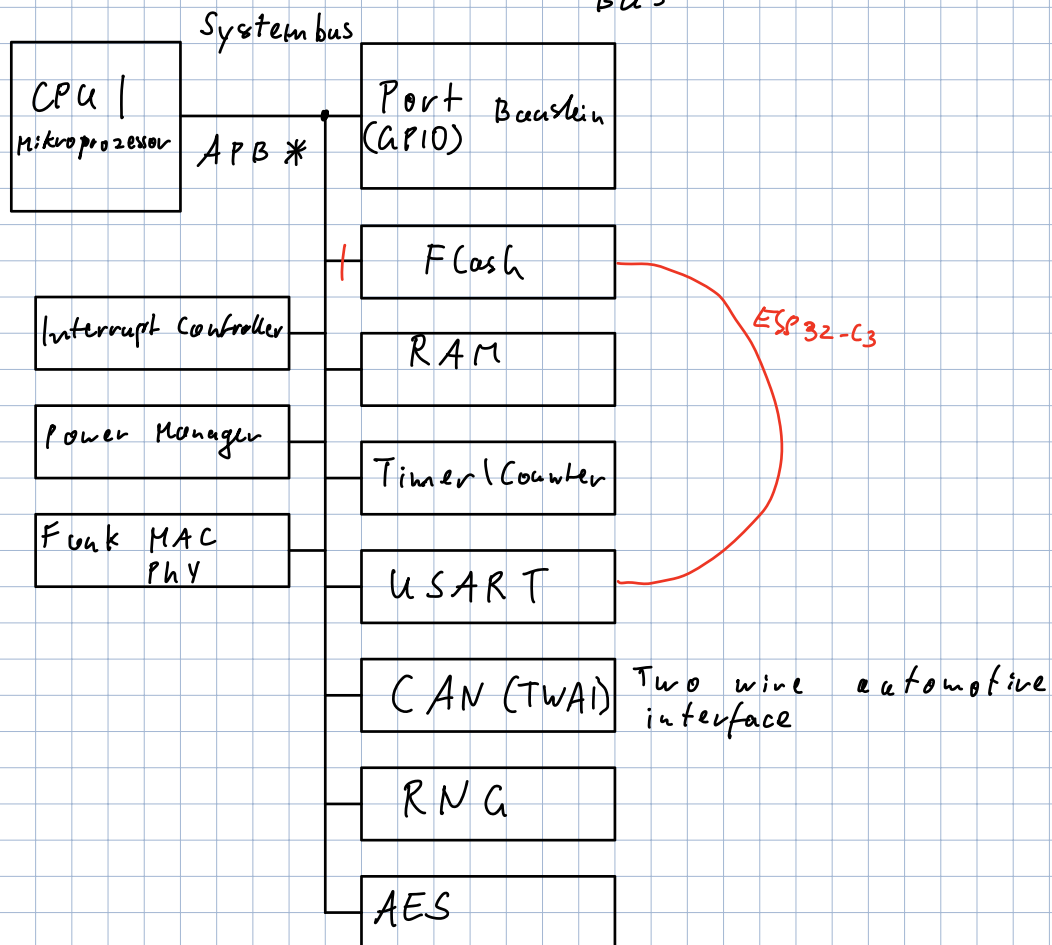


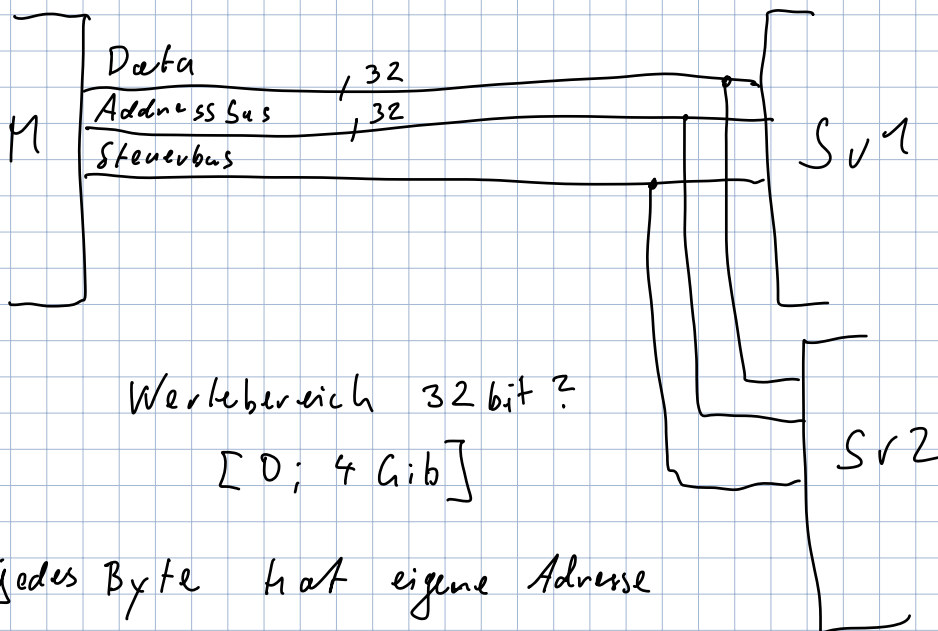
Cross-Platform Development

Bootloader / Basisprogramm auf ESP [Target System]

Mikrocontroller

Advanced Peripheral Bus

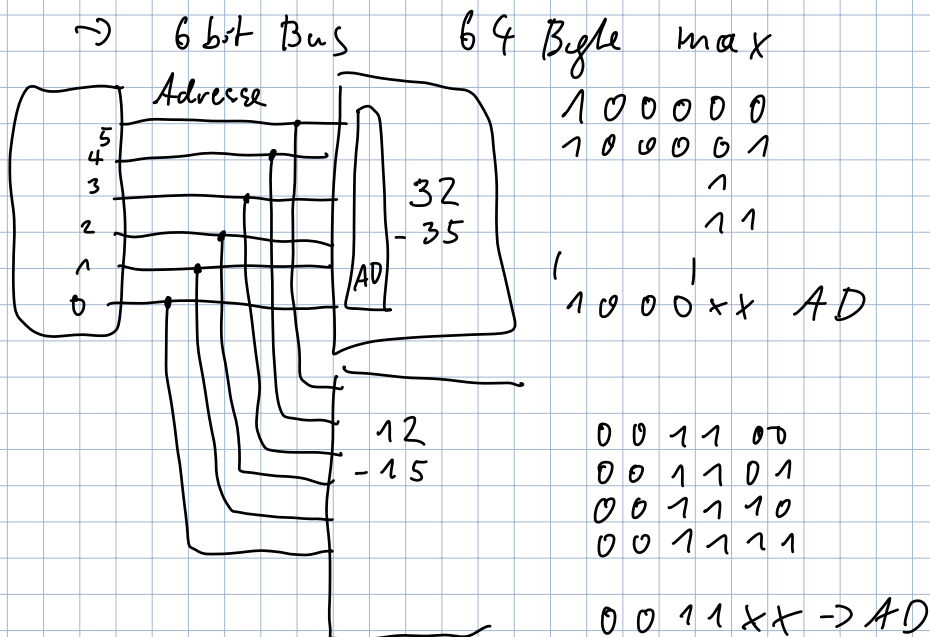


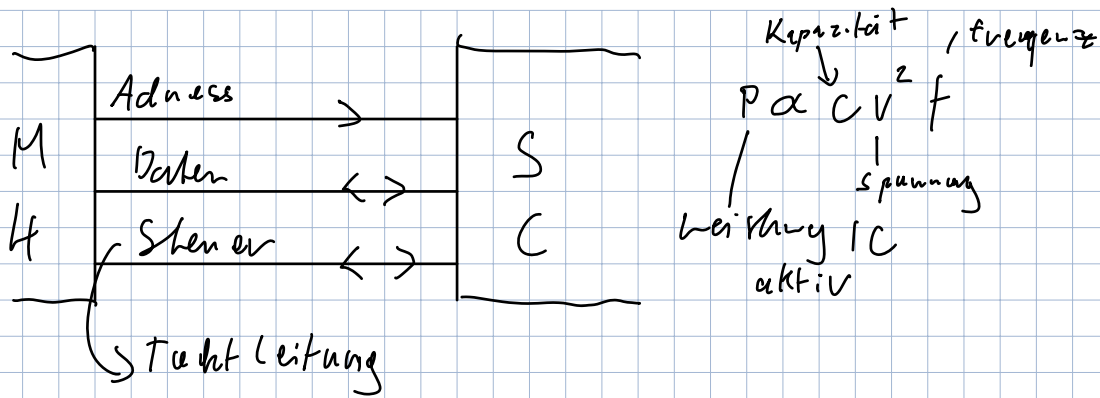


Wertebereich 32 bit?
 $[0; 4 \text{ Gib}]$

jedes Byte hat eigene Adresse
 $\Rightarrow 4 \text{ GB max}$

Computer Architecture Hennessey Patterson





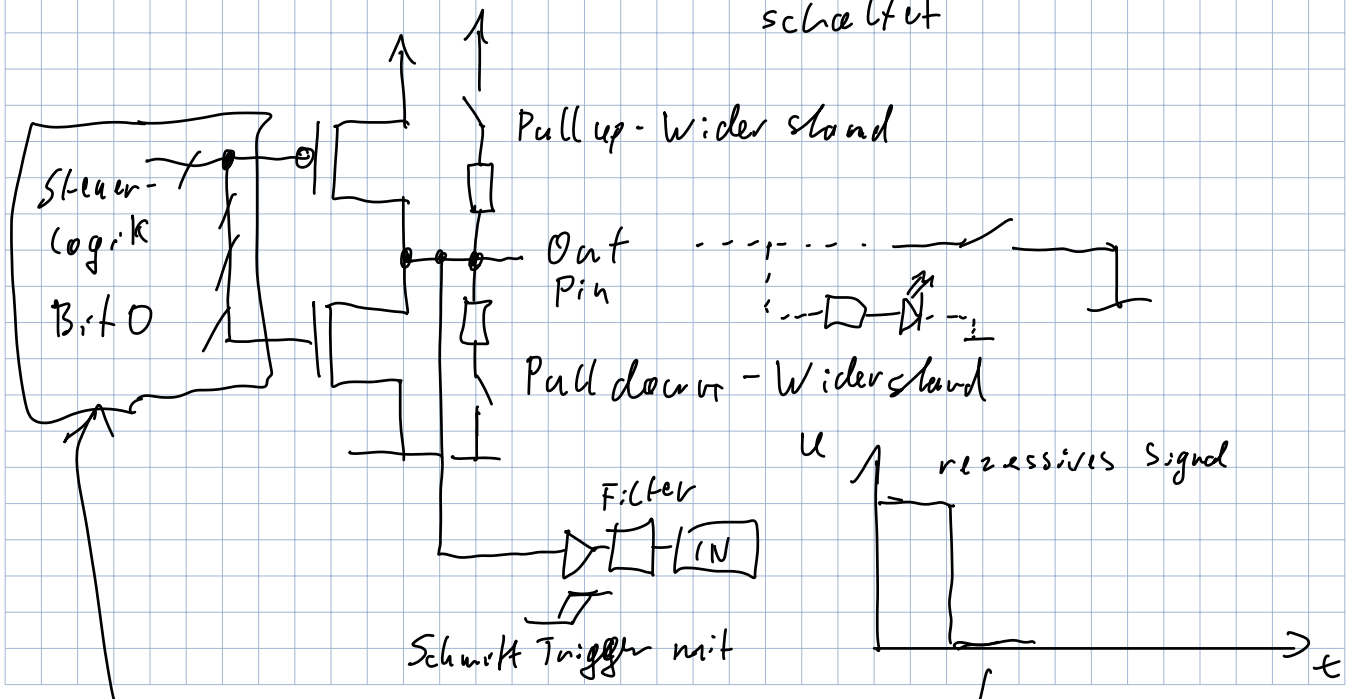
$r / \bar{w}$
 active / active
 High / Low

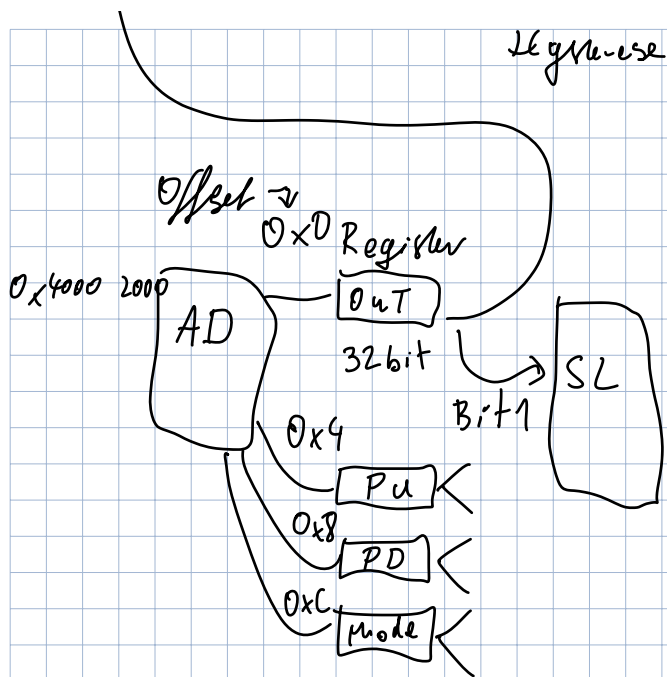
Port bzw. GPIO-Modul

General purpose

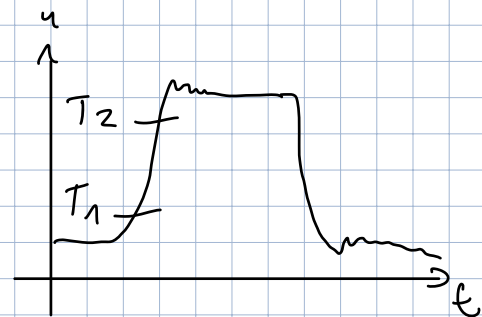
input-output $\rightarrow$ Digital

Push/Pull: beider Transistoren schalten
 Open-Drain: nur unterer Transistor schaltet





dominantes Signal



Glitch / Spike
Impuls
über Schwellenspannung

für PU-Reg

$0x40002004$

Memory Mapped I/O

```
uint32_t * pPup = (uint32_t*)0x40002004;
```

```
*pPup = 0x00000001;
```